Milestone 2 - Draft White Paper

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DSC 680

Topic

For this project, I will analyze a dataset on Crohn's Disease and Ulcerative Colitis patients to create a model that can predict if a flare will occur, defined as an elevated calprotectin level, based on gut bacterial composition.

Business Problem

Both Crohn's Disease and Ulcerative Colitis are chronic autoimmune diseases that cause elevated levels of inflammation in the body and damage the GI tract. Patients can remain in remission with treatment, however flares, which are periods of active inflammation, can occur and lead to damage. Predicting when a flare may occur would allow medical professionals to intervene earlier and try and prevent further damage.

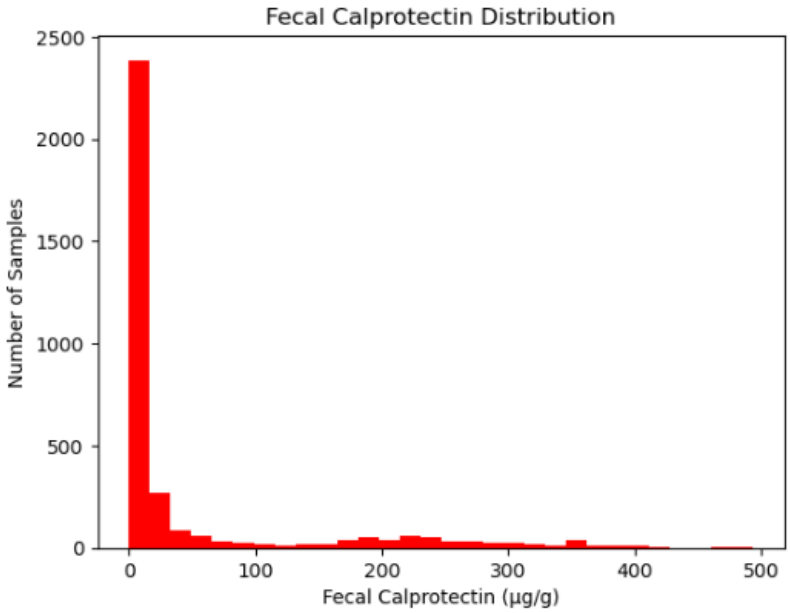
Background/History

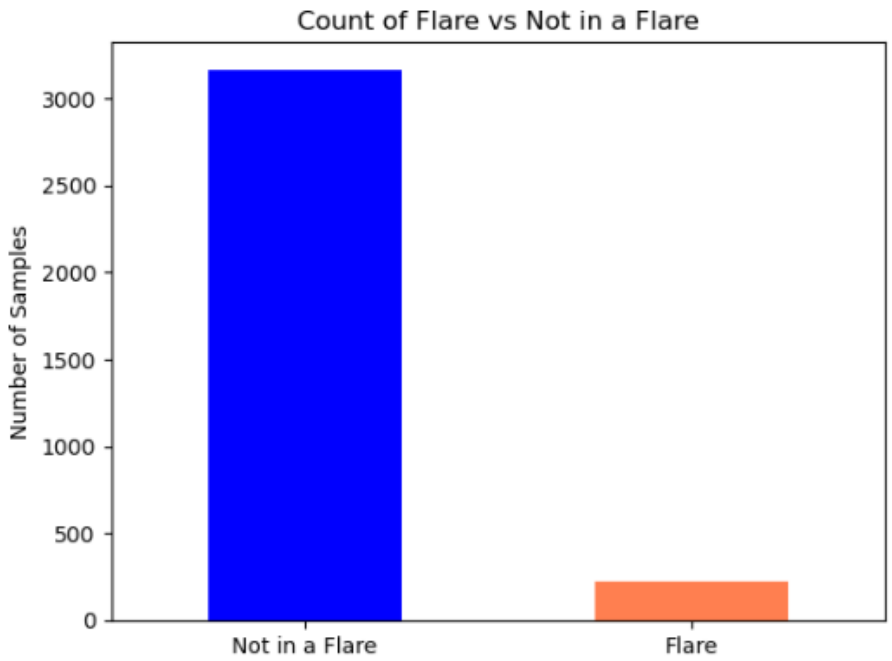
Crohn's disease affects the entire GI tract, while Ulcerative colitis affects mainly the colon. Neither condition has a cure, and symptoms are different for each patient. Patients with these diseases may experience flares which are defined as having active symptoms and inflammation (Crohn's & Colitis Foundation, 2026). Flares can also be identified using fecal calprotectin levels as well, which is a marker of inflammation in the intestines. These diseases can also increase in severity over time especially if the disease is not managed well and flares continue to occur (Crohn's & Colitis, 2023). Detecting flares early is important to prevent permanent damage due to excessive inflammation.

Data Explanation

The data that I am using for this project comes from a Kaggle dataset that curated data from the Integrative Human Microbiome Project (iHMP) which is a study that follows 132 individuals that have either Crohn's Disease, Ulcerative Colitis or are healthy for over a year to see how their gut microbiome changes during flares and remission (Qasim et al., 2022). This dataset contains information on various bacteria in the body along with fecal calprotectin levels, diagnosis, participant ID and the week of the study that the information was collected. With this data set I hope to be able to see how a participant's microbiome has changed throughout the study and to see how it may correlate with an elevated fecal calprotectin level and flare.

I changed null values to 0 in the "fecalcal" column and used fecal calprotectin levels to create a new variable named flare to show if the person was in a flare or not at that moment. I removed any extra columns that could cause data leakage during modeling like calprotectin levels. I also analyzed the data for a class imbalance based on the number of people experiencing flares as seen in the distribution and bar chart below.





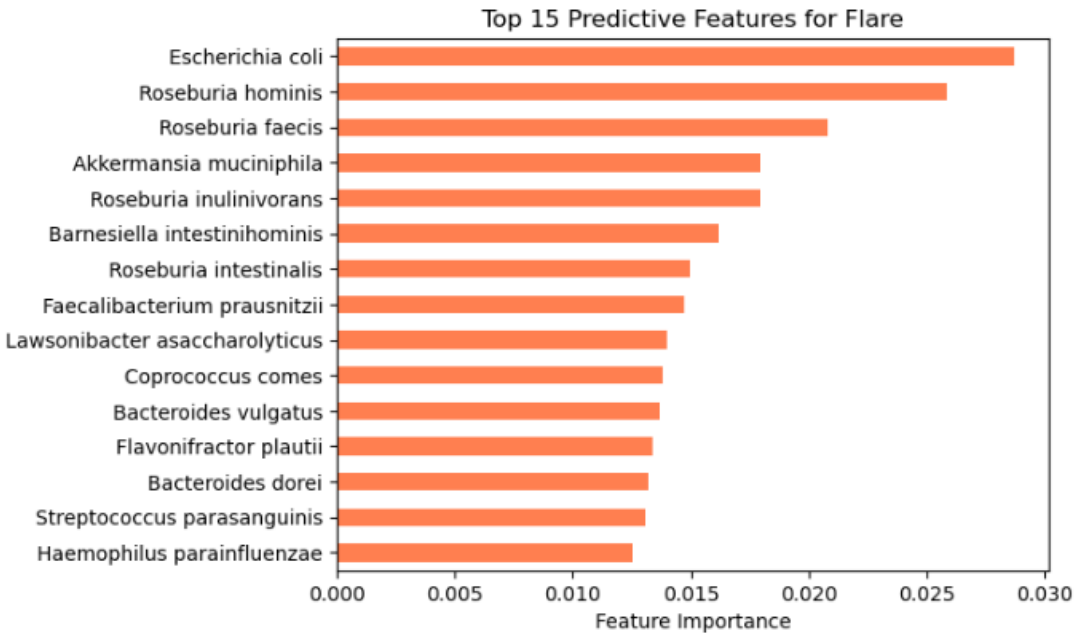
Methods

For this project, I used predictive modeling to determine the likelihood of a flare occurring using random forest modeling. I split the data into testing and training sets and evaluated how the model performed on unseen data. I also addressed the class imbalance by ensuring that participants with flares were properly represented in the model's training. The model performance was analyzed using metrics like accuracy, precision and recall. Feature importance analysis was used to determine which measures were the best indicators of high fecal calprotectin levels, which can help medical professionals determine which bacteria levels to look out for in their patients.

Analysis

The random forest model that I created based on this dataset has an accuracy of 99% without hyperparameter tuning. This result was obtained with adding "class_weight =

balanced" and "stratify=y" to make sure that the imbalance of patients in an active flare did not unintentionally bias the model. The model recall was 91%, the precision was 95% and the F1 score was 93% for flare cases. This indicates that the model did discover a large number of patients in active flares however it did miss a few. The recall is important in this scenario since missing patients who are in flares would lead to a delay in treatment and could cause issues. The F1 score shows that the model was able to find microbiome data that was strongly associated with flares and feature importance showed that some bacteria were more influential for predictions than others as seen below.



Conclusion

This analysis showed that microbiome data could likely be used to predict flares using a random forest model. The model was able to predict which patients were in flares strongly and had high accuracy, precision and recall indicating that there was a relationship between flares and gut microbiome data. Although the model does have limitations due to

sample size and a class imbalance, it could be a helpful addition to current methods used to determine if a patient is about to be in a flare and help with early intervention.

Assumptions

I assumed that the fecal calprotectin levels could be used to determine if someone was in a flare, since I cannot know for certain if they are experiencing symptoms, but a high fecal calprotectin number typically means that a patient has active inflammation. I also assumed that the null values in the "fecalcal" column were meant to be 0s since 0 was missing from this column.

Limitations

Limitations related to this project include that there are only 132 people involved in the study which limits the results and information that I can collect. A model built on a smaller sample size may not translate as well when being used in the general population.

Challenges/Issues

One challenge is the imbalance of the dataset. The number of people in flares, greater than 250 micrograms per gram of fecal calprotectin levels, is much smaller than the people who are considered to not be in a flare. This could cause an issue with the model and create bias if this model is used with data from the general public. There is also a lot of microbiome data in this dataset that could make feature importance complicated.

Future Uses/Additional Applications

This model could be tweaked to apply to other autoimmune diseases like Rheumatoid Arthritis that have different inflammation indicators. This model could also be used for diseases that are not autoimmune to see if there are any bacteria or other inflammation markers that may indicate that a person is about to develop symptoms.

Recommendations

I would recommend more work be done on this model to tune the parameters more or to collect more data that could be used to create a more accurate model. I also would recommend creating a new similar study to see how participants levels change within a year and see how well this model accurately predicts flares before being used fully in production by medical professionals.

Implementation Plan

To implement this model, more testing should be done on patients and data collected to make sure that the model can accurately predict if someone is about to enter a flare. Stool samples may be recommended be taken by patients more often to collect this data and to see where their gut microbiota levels are currently at. By having an idea of whether a flare may occur based on bacteria levels in the body, a doctor could tell a patient that indicates a potential flare what the plan is if they begin to develop any symptoms and nipping it in the bud before the flare gets worse. I would recommend initially using the model on a few patients who have less severe flares that can be tested or monitored easily.

Ethical Assessment

Some ethical considerations include that personally identifiable information should not be included since that will infringe on HIPAA laws. The information discovered in this model should also be used for research purposes or as clinical support, and not be the only factors used to diagnose or manage disease progression. The results are also only limited to the 132 participants in the study and may not be relevant for all IBD patients.

10 Questions the Audience May Ask

1. Why did you not choose a decision tree model?
2. How did you change the model so that the data was not overfit?
3. How can this model be updated to ensure it stays accurate over time?
4. What specific data needs to be collected to use this model in the future?
5. How did you mitigate ethical concerns?
6. Why is handling the class imbalance of the dataset important?
7. How did you know to define a flare as 250 micrograms per gram or higher?
8. Which features were most important in predicting flares?
9. Why is recall important in this situation?
10. How could this model be used to help medical professionals?

References

Crohn's & Colitis. (2023). *Crohn's disease severity and progression*.

https://www.crohnsandcolitis.com/crohns/severity-and-progression

Crohn's & Colitis Foundation. (2026). *About inflammatory bowel diseases*.

https://www.crohnscolitisfoundation.org/

Inflammatory Bowel Disease Multi-'omics Database (IBDMD). (n.d.). *IBDMDB*.

https://ibdmdb.org/

Qasim, H.*, et al.* (2022). *Human gut microbiome atlas (HMP2) [Data set]. Kaggle*.

https://www.kaggle.com/datasets/qasimhu/human-gut-microbiome-atlas-hmp2

Appendix A: Code used for Analysis

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Calprotectin Levels Correlation with Crohn's Disease and Colitis Flares

In [148…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [150…

```
df = pd.read_csv("hmp2_ibd_metagenomics_atlas_20260219_121629.csv")
```

In [152…

```
df.head()
```

Out[152…

	External ID	Participant ID	week_num	diagnosis	fecalcal	Bacteroides vulgatus	Bacteroides uniformis	thetai
0	MSMA26EH	M2069	39	Ulcerative Colitis	493.7	0.0000	0.00000	
1	MSMA26EH	M2069	39	Ulcerative Colitis	493.7	0.0000	0.00000	
2	MSMA26EH	M2069	39	Ulcerative Colitis	493.7	0.0000	0.00000	
3	MSMA26EH	M2069	39	Ulcerative Colitis	493.7	0.0000	0.00000	
4	HSMA33LB	H4044	16	Ulcerative Colitis	471.3	27.3852	4.15501	

5 rows × 571 columns

In [154…

```
#check for any null values
df.isnull().sum()
```

Out[154…

```
External ID                          0
Participant ID                       0
week_num                             0
diagnosis                            0
fecalcal                          2027
                                   ...
Actinobaculum sp oral taxon 183      0
Lachnoclostridium sp An298           0
Haemophilus haemolyticus             0
Enterococcus dispar                  0
Atopobium minutum                    0
Length: 571, dtype: int64
```

In [156…

```
#show columns with nulls
null_columns = df.columns[df.isnull().any()]
print(null_columns)
```

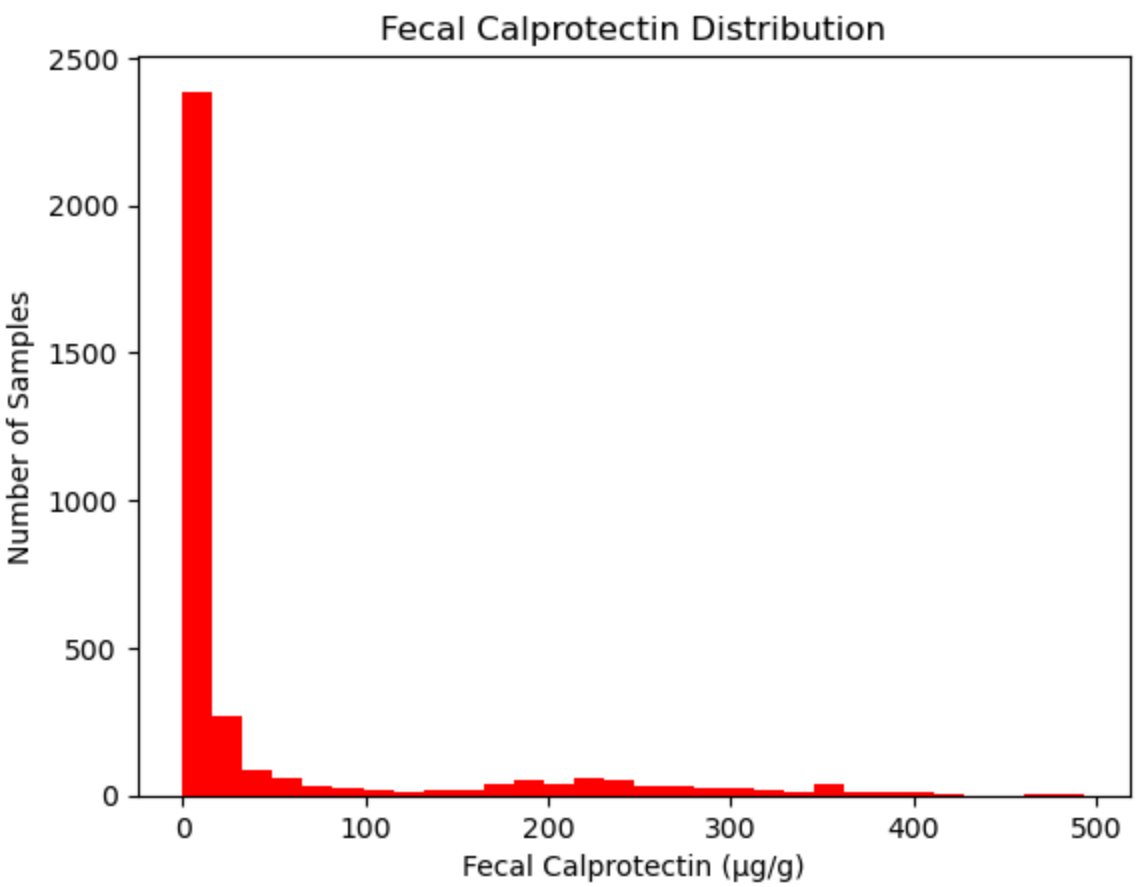
```
Index(['fecalcal'], dtype='object')
```

In [158…

```
#change null fecalcal levels to 0
df['fecalcal'] = df['fecalcal'].fillna(0)
```

In [160…

```
#check to see the distribution of fecalcal level in the dataset
plt.hist(df['fecalcal'], bins=30, color='red')
plt.xlabel('Fecal Calprotectin (µg/g)')
plt.ylabel('Number of Samples')
plt.title('Fecal Calprotectin Distribution')
plt.show()
```



First I will need to define what a flare is, which is a level of 250 micrograms per gram or higher (an elevated fecal calprotectin level) and to add this label to the dataset as a number of 1 or 0.

In [163…

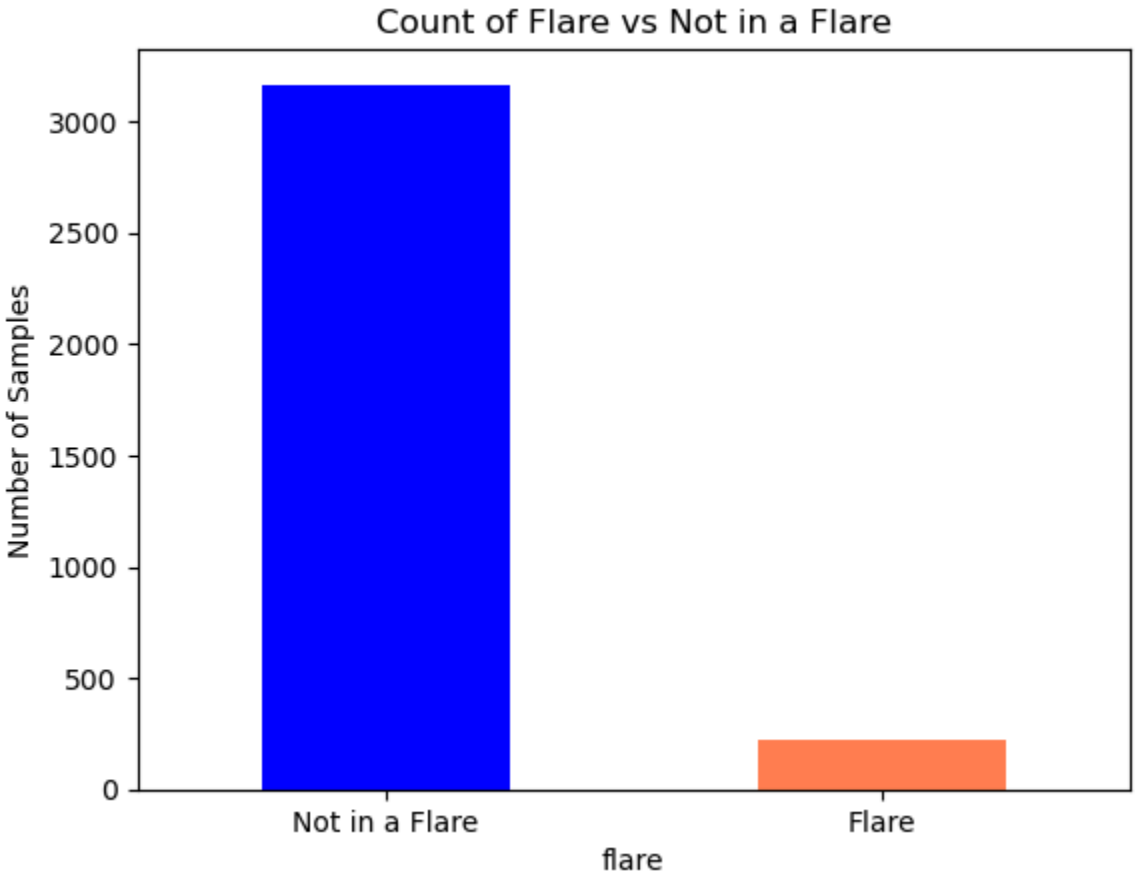
```
#add a column to state whether the person is in a flare or not
df['flare'] = (df['fecalcal'] > 250).astype(int)

df['flare'].value_counts()
```

Out[163…

```
flare
0    3165
1     222
Name: count, dtype: int64
```

In [165…

```
df['flare'].value_counts().plot(kind='bar', color=['Blue', 'Coral'])
plt.xticks([0,1], ['Not in a Flare', 'Flare'], rotation=0)
plt.ylabel('Number of Samples')
plt.title('Count of Flare vs Not in a Flare')
plt.show()
```



In [167…

```
df.head()
```

Out[167…

	External ID	Participant ID	week_num	diagnosis	fecalcal	Bacteroides vulgatus	Bacteroides uniformis	thetai
0	MSMA26EH	M2069	39	Ulcerative Colitis	493.7	0.0000	0.00000	
1	MSMA26EH	M2069	39	Ulcerative Colitis	493.7	0.0000	0.00000	
2	MSMA26EH	M2069	39	Ulcerative Colitis	493.7	0.0000	0.00000	
3	MSMA26EH	M2069	39	Ulcerative Colitis	493.7	0.0000	0.00000	
4	HSMA33LB	H4044	16	Ulcerative Colitis	471.3	27.3852	4.15501	

5 rows × 572 columns

The dataset is a bit imbalanced which will affect how i evaluate my model later on and is important to note.

In [170…

```
#drop columns that are not important and the target flare column as well from X
X = df.drop(columns=[
    'External ID',
    'Participant ID',
    'fecalcal',
    'diagnosis',
    'flare'
])

y = df['flare']
```

Next I will create testing and training splits in order to train the model and measure it's effectiveness.

In [173…

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    #adding stratify to see if it will help with the imbalance
    stratify=y
)
```

In [175…

```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(
    n_estimators=100,
    random_state=42,
```

```
    class_weight='balanced'
)

rf.fit(X_train, y_train)
```

Out[175…

```
RandomForestClassifier
RandomForestClassifier(class_weight='balanced', random_state=42)
```

In [176…

```
y_pred = rf.predict(X_test)
```

In [179…

```
importances = pd.Series(rf.feature_importances_, index=X.columns)
top_features = importances.sort_values(ascending=False).head(15)

print(top_features)
```

```
Escherichia coli                  0.028733
Roseburia hominis                 0.025857
Roseburia faecis                  0.020781
Akkermansia muciniphila           0.017967
Roseburia inulinivorans           0.017931
Barnesiella intestinihominis      0.016154
Roseburia intestinalis            0.014960
Faecalibacterium prausnitzii      0.014695
Lawsonibacter asaccharolyticus    0.014009
Coprococcus comes                 0.013807
Bacteroides vulgatus              0.013711
Flavonifractor plautii            0.013387
Bacteroides dorei                 0.013212
Streptococcus parasanguinis       0.013060
Haemophilus parainfluenzae        0.012558
dtype: float64
```

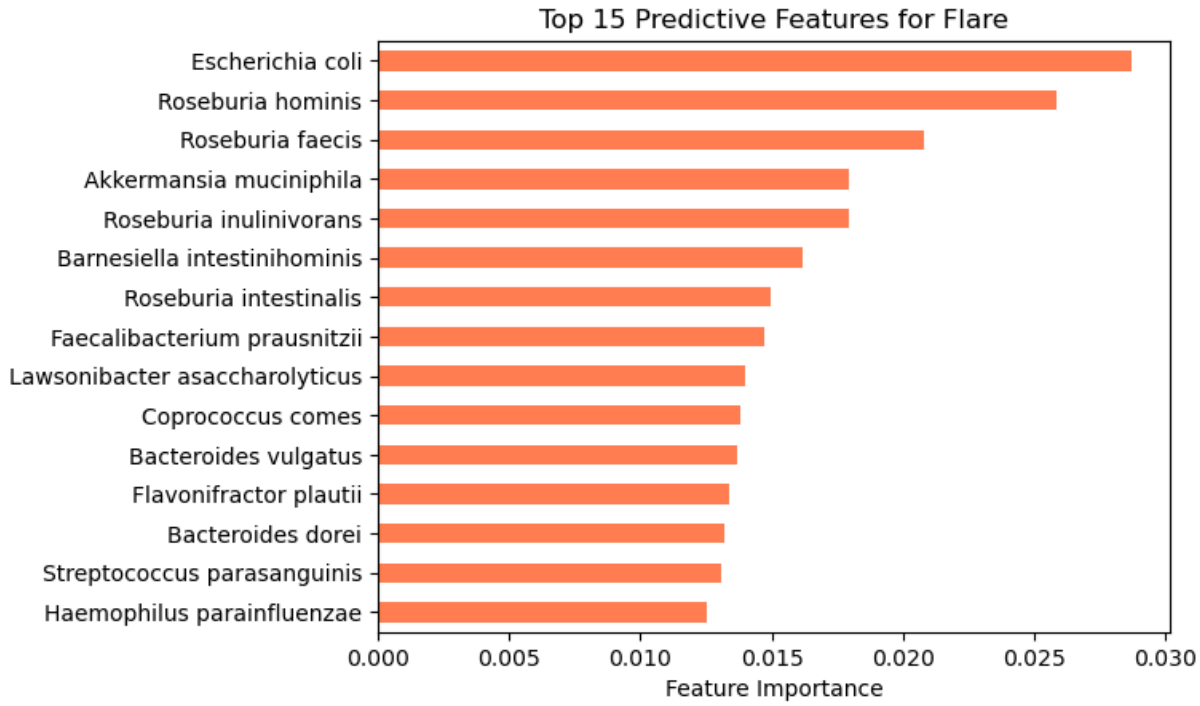
In [181…

```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.99      1.00      1.00       634
           1       0.95      0.91      0.93        44

    accuracy                           0.99       678
   macro avg       0.97      0.95      0.96       678
weighted avg       0.99      0.99      0.99       678
```

In [183…

```
import matplotlib.pyplot as plt

top_features.sort_values().plot(kind='barh', color='coral')
plt.xlabel('Feature Importance')
plt.title('Top 15 Predictive Features for Flare')
plt.show()
#µg/g is the unit
```


Top 15 Predictive Features for Flare
Escherichia coli
Roseburia hominis
Roseburia faecis
Akkermansia muciniphila
Roseburia inulinivorans
Barnesiella intestinihominis
Roseburia intestinalis
Faecalibacterium prausnitzii
Lawsonibacter asaccharolyticus
Coprococcus comes
Bacteroides vulgatus
Flavonifractor plautii
Bacteroides dorei
Streptococcus parasanguinis
Haemophilus parainfluenzae
0.000
0.005
0.010
0.015
0.020
0.025
0.030
Feature Importance

```
In [ ]:
```